Experiment 4

Question

Perform basic Image Handling and processing operations on the image. Read an image in Python and dilate an image using the Dilate function.

Aim

To read an image using Python and OpenCV and apply the Dilation operation to enlarge the bright regions of the image.

Procedure

1. Import the OpenCV (cv2) and NumPy libraries.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Create a kernel (structuring element) using NumPy.
6. Apply the dilation operation using cv2.dilate().
7. Display the dilated image.
8. Save the dilated image.

Code

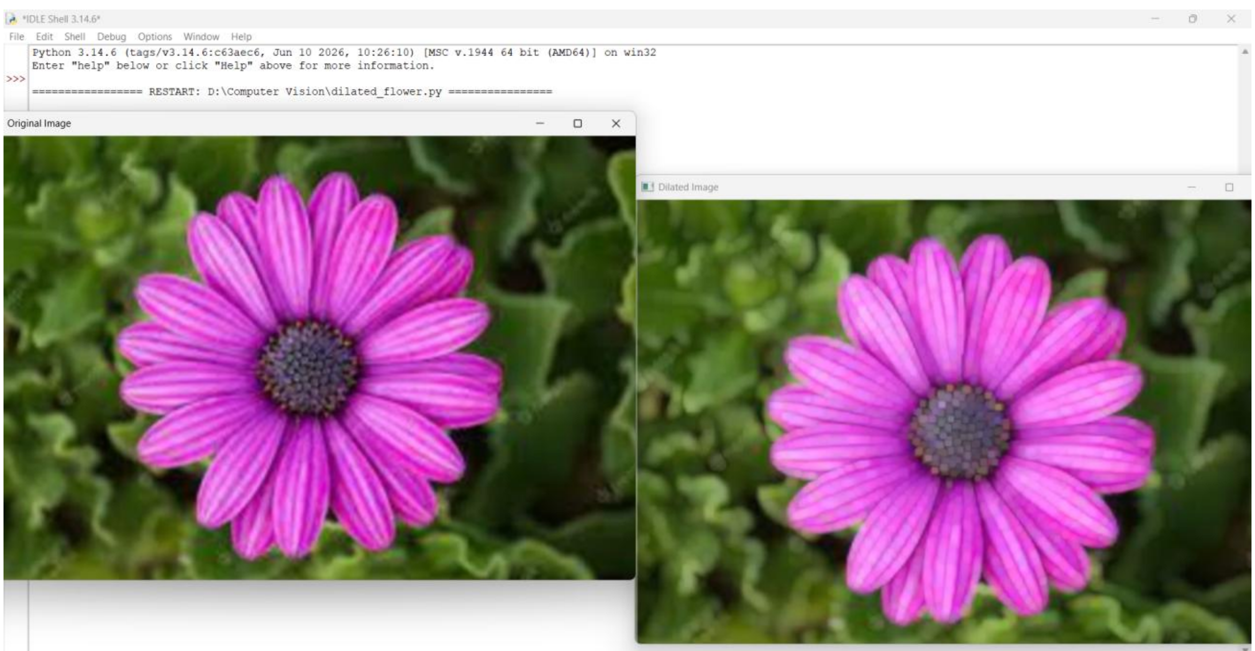
```
import cv2
import numpy as np

# Read image
image = cv2.imread("flower.jpg")

# Check whether image is loaded
if image is None:
    print("Image not found!")
else:
    # Display original image
    cv2.imshow("Original Image", image)

    # Create kernel
    kernel = np.ones((5,5), np.uint8)

    # Apply dilation
    dilated = cv2.dilate(image, kernel, iterations=1)

    # Display dilated image
    cv2.imshow("Dilated Image", dilated)

    # Save output image
    cv2.imwrite("dilated_flower.jpg", dilated)
```

Input

Input Image: flower.jpg

Output

1. **Original Image**
    - Displays the original flower image.
2. **Dilated Image**
    - Displays the image after applying the dilation operation.
3. **Saved File**
    - dilated_flower.jpg is created in the project folder.



Result

The image was successfully read using OpenCV, the dilation operation was applied, and the dilated image was displayed and saved successfully.